

CSCI 5822

Probabilistic and Causal Modeling in Computer Science

FORMAT	CATEGORY	CREDITS
On-Campus	Elective	3
SECTION	BEST ROLE MATCH	AVG ALIGNMENT
Section 4 – Elective (On-Campus)	Data Scientist (100.0 / 100)	69.4 / 100

69	97	+44
CURRENT AVG SCORE across 6 job roles	POTENTIAL SCORE with recommended additions	MAX SCORE GAIN best single-role improvement

CURRENT COURSE DESCRIPTION

Introduces a set of modeling techniques that have become a mainstay of modern artificial intelligence and machine learning research. These techniques combine graphical models, Bayesian analysis, and multivariate statistics for probabilistic and causal inference and for interpreting the statistical structure of large data sets. Applications include healthcare, economics, marketing, social sciences, and more.

INDUSTRY ALIGNMENT BY JOB ROLE

JOB ROLE	CURRENT SCORE	PROJECTED SCORE	POINT GAIN	PRIORITY
Data Scientist	100.0	100.0	+0.0	LOW
Machine Learning Engineer	75.8	100.0	+24.2	MEDIUM
ML Engineer	67.5	91.7	+24.2	MEDIUM
AI Engineer	66.7	100.0	+33.3	HIGH
Applied Scientist	61.0	100.0	+39.0	HIGH
MLOps Engineer	45.1	88.6	+43.5	HIGH

RECOMMENDED TOPICS TO ADD

Keywords highlighted in gold (example) appear as gaps across 3 or more job roles.

Deep Learning 2 of 6 roles

deep training

Feature Engineering 1 of 6 roles

preprocessing

PRIORITY RECOMMENDATIONS FOR INSTRUCTORS

1 **MEDIUM** **Consider Deep Learning content** — relevant to 2 industry roles. Could be introduced via case studies or optional readings. Key terms: deep, training.

2 **LOW** **Optional: Feature Engineering** — would broaden appeal to one additional role. Candidate terms: preprocessing.

GAP KEYWORDS BY ROLE

Machine Learning Engineer

75.8 → 100.0 (+24.2 pts)

web tools exploratory design science model information develop

ML Engineer

67.5 → 91.7 (+24.2 pts)

model problems tools detection

AI Engineer

66.7 → 100.0 (+33.3 pts)

following model problems detection information design tools deep

Applied Scientist

61.0 → 100.0 (+39.0 pts)

biology computational problems industry design real science testing

MLOps Engineer

45.1 → 88.6 (+43.5 pts)

detection model text preprocessing image training tools

Methodology: Scores (0–100) reflect TF-IDF cosine similarity between course catalog descriptions and a corpus of 100 LinkedIn data science job postings (collected June 2026, filtered to postings with ≥200 applicants). Gap keywords are terms that appear frequently in job postings for a given role but are absent or underrepresented in the current course description. "Projected score" shows the estimated alignment after integrating the top 5 suggested keywords. Analysis was conducted by the CU Boulder MSDS Curriculum-Industry Alignment project team.